AI-POWERED SAAS FOR SPECIALTY AGRICULTURE

Oyster360

From experience to intelligence.

Multi-tenant farm management & intelligence for commercial oyster mushroom cultivation.

THE PROBLEM

Mushroom farms run on paper, memory and luck

- **Fragmented records.** Batches, environment readings and harvests live in notebooks and Excel — no history, no analysis.
- **Late problem detection.** Contamination (Trichoderma, cobweb) is often found only after it spreads — 15–25% of batches lost.
- **Inconsistent yields.** 600–900 g per bag with high variation; nobody knows which strain, recipe or climate drove success.
- **Knowledge trapped in people.** Success depends on a few experienced growers; scaling stalls and training takes months.

Result: thin margins, wasted substrate, and farms that cannot scale past ~5,000 bags per manager.

THE SOLUTION

One platform for the whole farm, with AI on top

Oyster360 is a multi-tenant SaaS that digitizes the entire oyster mushroom operation — batch lifecycle (preparation → inoculation → colonization → fruiting → harvest), rooms and grow spaces, strain catalogue, versioned substrate recipes, environment records (temperature / humidity / CO■), inventory & purchasing, harvest grading and revenue — then layers AI decision support on each farm's own data.

24	**28**	**28**	**4**
API modules	service modules	app pages	roles / RBAC

AI TECHNOLOGY

Four AI engines, grounded in farm data

Cultivation Assistant (NLP + GenAI)

Natural-language Q&A; grounded in live batch, environment and growth data. Multi-provider LLM layer (OpenAI / Gemini / local) with a deterministic rule-based fallback so the assistant always works.

Knowledge RAG

Farms upload SOPs and notes; documents are chunked, stored and retrieved per-user to augment every answer — farm knowledge becomes searchable and reusable.

Image Inspection (Computer Vision)

Photos of bags/substrate are analyzed for health score, contamination probability and growth stage, with findings and concrete corrective actions.

Yield Prediction (ML)

Feature-based model predicts kg per batch, confidence score and expected harvest date from strain, recipe and environment history.

DIFFERENTIATION

What makes Oyster360 unique

- **Domain-specific, not generic agri-software.** Built around oyster mushroom biology: substrate recipes, colonization stages, flush cycles, harvest grading.
- **AI-native from day one.** Not dashboards-only — assistant, RAG, vision inspection and yield prediction are core workflows, running as tenant-scoped background jobs.
- **Works with or without an AI key.** Deterministic rule-based engines guarantee reliable advice offline; LLM providers upgrade it when available.
- **Production-grade SaaS.** Tenant isolation, RBAC, MFA/TOTP, Stripe billing with signed webhooks, GDPR export/delete, Docker + CI/CD.

IMPACT

Measured impact targets

+15–25%

yield per bag (650–750 → 780–850 g)

–50–60%

contamination losses (15–25% → 6–10%)

3×

bags per manager (5k → 15k+)

80%

faster problem resolution

60%

faster staff onboarding (3–6 mo → 2–4 wks)

100%

of decisions backed by farm data

PRODUCT

What farm teams use every day

Cultivation & quality

- Batch lifecycle & stage tracking
- Rooms, strains & recipe versions
- Environment logging (T / RH / CO_2)
- AI assistant chat, grounded in live data
- Photo inspection & contamination findings

Operations & business

- Yield forecasts & harvest planning
- Harvest grading & quality scores
- Inventory, suppliers & purchase orders
- Farm dashboards & KPIs
- Subscription billing & admin analytics

MARKET

Who we serve

- **Medium to large oyster mushroom farms** (1,000+ bags per cycle) seeking consistent yield and lower contamination.
- **Multi-site mushroom companies** that need standardized processes and cross-site visibility.
- **Agricultural consultants & researchers** who need structured cultivation data and traceable outcomes.
- Oyster mushrooms are one of the fastest-growing specialty food segments — low entry capital, high spoilage risk, and almost no dedicated software today.

COMMERCIAL

Business model

- **SaaS subscriptions** billed through Stripe — server-controlled pricing, verified webhooks, idempotent subscription sync, cancellation at period end.
- **Multi-tenant isolation:** every farm is an organization with enforced data boundaries — one deployment serves many customers.
- **Expansion paths:** per-seat and per-bag tiers, sensor integrations, consultancy analytics, and white-label deployments.

Billing infrastructure is implemented end-to-end; the platform is commercially deployable today.

TECHNOLOGY

Built for production

- **Frontend:** Next.js 16, React 19 + TypeScript, Tailwind 4, TanStack Query, Chart.js
- **Backend:** FastAPI, Pydantic v2, SQLAlchemy 2, Alembic, PostgreSQL 16 + pgvector
- **Async:** Celery workers + Beat on Redis for AI jobs, emails, reports, cleanup
- **Security:** JWT + rotating refresh tokens, Argon2, TOTP MFA, RBAC, rate limits, audit logs
- **Compliance:** GDPR export & deletion, data retention services
- **Ops:** Docker multi-stage images, Compose topology, GitHub Actions CI, Trivy scans, tests

WHERE WE ARE

Status & roadmap

Shipped

- Multi-tenant isolation & security suite
- Full cultivation + inventory + purchasing workflows
- AI assistant, RAG, vision inspection, yield prediction
- Stripe billing & SaaS analytics
- Docker deployment & CI/CD

Next

- Real-time sensor ingestion (IoT)
- Embedding-based RAG over pgvector
- Fine-tuned contamination vision model
- Mobile field app & offline mode

From experience to intelligence.

Oyster360 gives every oyster mushroom farm the operating system — and the AI — to grow more with

Try the demo · github.com/Inkithai/Oyster360